

Concepts of Nano-Programming (Self-Learning)

Q. Discuss the concepts of Nano Programming in computer architecture. What are Nano Instructions and Nano Memory, and how do they optimize microprogramming?

📌 Definition to Remember

Nano Programming is a secondary level of control abstraction that compresses large microprograms. By splitting memory into **Micro Memory** (pointers) and **Nano Memory** (actual control signals), it eliminates redundant control signal patterns, saving physical chip space and hardware cost.

1. The Problem with Standard Microprogramming

In CISC processors, microprograms are very large. Each microinstruction contains a long bit pattern (64-128 bits) directly representing control signals. Statistically, across thousands of microinstructions, many duplicate the exact same control bit patterns. Storing these redundant, long strings repeatedly wastes expensive Control Memory (ROM).

2. The Nano Programming Solution (Two-Level Hierarchy)

Nano programming eliminates redundancy using a two-level memory setup:

1. **Micro Memory:** Contains shorter microinstructions. Instead of holding control signals, they act as **pointers (addresses)** to the Nano Memory.
2. **Nano Memory:** Contains the actual, long control signal patterns (**Nano Instructions**). Duplicates are eliminated; unique patterns are stored only once.

3. How It Works

1. Sequencer generates a micro-address.
2. CPU reads the short microinstruction from **Micro Memory**.
3. This microinstruction provides an address to the **Nano Memory**.
4. CPU fetches the **Nano Instruction** (the long control word) and applies it to the hardware.

4. Memory Structure Comparison

- **Nano Memory:** Small number of entries (duplicates removed), but very **wide** (e.g., 100 bits) to hold control signals.
- **Micro Memory:** Large number of entries (one for every step), but very **narrow** (e.g., 10 bits), holding only pointers.

5. Advantages and Disadvantages

- **Advantage (Space Saving):** Massively reduces total ROM bits required on the chip, lowering manufacturing cost.
- **Disadvantage (Speed Penalty):** Generating signals requires **two memory accesses** (Micro then Nano) instead of one, slightly increasing execution time.

★ Must-Write Points (for 10 marks)

1. Standard microprogramming wastes ROM by storing duplicate control signal patterns.
2. **Nano Programming** optimizes this via a Two-Level Control Memory Hierarchy.
3. **Micro Memory** stores short pointers (addresses).
4. **Nano Memory** stores the actual, wide **Nano Instructions** (control signals).
5. Duplicate control patterns are stored only once in Nano Memory.
6. **Advantage:** Saves massive amounts of physical ROM space on the CPU chip.
7. **Disadvantage:** Causes a slight speed penalty because it requires two memory lookups instead of one.

⚡ Quick Recall

Nano Programming → Solves ROM waste (duplicate patterns) → 2 Levels: Micro Memory (stores pointers, narrow) + Nano Memory (stores actual signals, wide) → Pros: Saves space/cost → Cons: 2 lookups = slower